

The Sentiment-Head Loss: Weighted Huber + CCC

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The problem this loss solves

Evaluation showed the production model **under-polarizes**: it predicts sentiment magnitudes about a third of the true value (predicted std 0.167 vs. label std 0.275) and essentially never outputs a “very negative” score. A post-hoc calibration test proved this is **regression attenuation** — the mathematically correct hedge for a model with limited ranking accuracy — and the old training recipe (MSE + (1 − Pearson)) actively encourages it. The new `ccc_huber` loss is designed to remove that incentive.

The loss has three additive terms:

$$L = w_{\text{huber}} \cdot \text{weightedHuber} + w_{\text{ccc}} \cdot (1 - \text{CCC}) + w_{\text{sign}} \cdot \text{signPenalty}$$

1. Huber loss — robust regression

Huber loss blends squared error (L2/MSE) and absolute error (L1/MAE). For an error $e = \hat{y} - y$ and threshold δ :

$$\text{Huber}_{\delta}(e) = \begin{cases} \frac{1}{2} e^2 & \text{if } |e| \leq \delta \quad (\text{MSE-like: smooth, precise near 0}) \\ \delta(|e| - \frac{1}{2}\delta) & \text{if } |e| > \delta \quad (\text{MAE-like: linear, gradient capped at } \delta) \end{cases}$$

We use $\delta = 0.1$.

Why Huber, not MSE. MSE squares the error, so a single badly-wrong label produces an outsized gradient. Our labels are LLM-generated and noisiest at the **extremes**, so under MSE one mislabeled -0.9 could yank the model. Huber **caps the gradient magnitude** beyond δ : large errors are still penalized, but linearly, so noisy extremes cannot dominate. We keep MSE’s precision near zero and MAE’s outlier-robustness.

Weighted Huber (what we actually use) scales each entity’s term by

$$w_i = \min(1 + 3|y_i|, 3).$$

A neutral entity gets weight ≈ 1 ; a strong entity ($|y_i| = 0.7$) gets ≈ 3 . Because roughly 59% of the labels are neutral, an **unweighted** loss is minimized by predicting ≈ 0 everywhere (mean-collapse). Up-weighting the strong-sentiment entities forces the model to commit to a magnitude.

2. CCC — Concordance Correlation Coefficient

CCC measures agreement with the **identity line** ($\hat{y} = y$), not merely correlation along *some* line:

$$\text{CCC} = \frac{2 \text{cov}(\hat{y}, y)}{\text{var}(\hat{y}) + \text{var}(y) + (\mu_{\hat{y}} - \mu_y)^2} = \frac{2 \rho \sigma_{\hat{y}} \sigma_y}{\sigma_{\hat{y}}^2 + \sigma_y^2 + (\mu_{\hat{y}} - \mu_y)^2},$$

where ρ is Pearson's correlation. The denominator stacks three penalties: prediction variance, target variance, and the squared gap between the means.

Implementation (population variance; 10^{-8} guards division by zero):

```
vp = pred - pred.mean();  vy = y - y.mean()
cov = (vp * vy).mean()
ccc = 2*cov / (pred.var() + y.var() + (pred.mean() - y.mean())**2 + 1e-8)
```

Why CCC and not Pearson — the crux. Pearson is **scale- and location-invariant**. A timid model predicting exactly half the true magnitude ($\hat{y} = 0.5 y$) scores Pearson = 1.0 — *perfect*. Pearson literally cannot see under-polarization. CCC **can**: shrinking the predictions shrinks cov in the numerator while the denominator stays large, so CCC falls. Maximizing CCC (i.e. minimizing $1 - \text{CCC}$) forces the model to match the labels' **spread and mean** — to stop hedging toward zero. This term directly punishes the magnitude compression the evaluation diagnosed.

3. Sign penalty

An asymmetric hinge applied only to strong-sentiment entities ($|y_i| \geq 0.4$), penalizing predictions on the **wrong side** of zero:

$$\text{signPenalty} = \text{mean}_{|y_i| \geq 0.4} \text{ReLU}(-\hat{y}_i \cdot y_i).$$

The product $\hat{y}_i \cdot y_i$ is negative exactly when prediction and label disagree in sign; ReLU keeps only those cases. Getting the *direction* wrong on a lawsuit or an earnings beat is worse than missing its magnitude, so this term guards the cases that matter most.

4. How the terms work together

			failure
			mode
		what	if
		it	used
term	scope	guarantee	alone
weighted	pointwise	individual	still
Huber	(each	accu-	drifts
	$\hat{y}_i$	racy,	to
	vs. its	robustly	the
	y_i)		mean
			on
			neutral-
			heavy
			data
1 –	distributional	wouldn't	
CCC	(whole	spread	pin
	batch	+	indi-
	vs. labels)	alignment	ind-
			ual
			predictions
sign	strong	no	ignores
penalty	en-	di-	mag-
	ti-	rec-	ni-
	ties	tion	tude
		flips	entirely
		where	
		it	
		counts	

The two main terms are complementary: Huber keeps individual predictions accurate, while CCC keeps the overall distribution from collapsing toward neutral. Contrast the **old** recipe, $\text{MSE} + (1 - \text{Pearson})$: a mean-collapsing term plus a scale-*blind* term — precisely why the current model is timid.

5. Empirical confirmation

Retraining only the sentiment head (frozen encoder) with this loss on the new decisive labels moved the held-out **test** Pearson from $0.485 \rightarrow 0.643$ and the predicted/label std ratio from $0.57 \rightarrow 1.04$ — i.e. the under-polarization is gone — while sign-flips *fell* ($0.215 \rightarrow 0.142$) and the extreme-bucket F1 scores rose from ≈ 0 . The loss redesign did what the diagnosis predicted.